Analysis of Large-Scale Multi-Stage All-Optical Packet Switching Routers

Q. Xu, H. Rastegarfar, Y. Ben M'Sallem, A. Leon-Garcia, S. LaRochelle, and L. A. Rusch

***Abstract*—All-optical packet switching can overcome limitations of electronic switches in terms of power consumption, speed, cost, and footprint. Switch architectures combining wavelength converters and fiber delay lines provide tunable routing and contention resolution when used with an $N \times N$ arrayed waveguide grating (AWG), a key passive optical component to bypass electronic processing limitations. An AWG passively routes either single or multiple input port wavelengths to output ports. A single wavelength per port strategy reduces crosstalk within the AWG, but drastically increases the dimensionality of the device. AWG design constraints due to bandwidth limitations and fabrication processes limit the port number for the foreseeable future to under 100. In order to scale optical switches to emerging network requirements, we must use multiple wavelengths per port. In this paper, we examine several optical router architectures for data center applications using multiple wavelengths per port, and quantify the physical layer impairments. We consider not only the AWG crosstalk, but also *Q*-factor degradation caused by the multiple wavelength conversions occurring when a packet is buffered for contention resolution. We present the results as a function of the number of recirculations for on–off-keying (OOK) signal formats. While previous work has addressed this issue in terms of accumulated loss, we focus on accumulated intensity noise due to crosstalk and amplified spontaneous emission (ASE). We compare the routing performance of each architecture, and we point out that the AWG crosstalk and accumulated ASE noise during packet recirculation are both critical to the routing performance.**

***Index Terms*—Arrayed waveguide grating; Optical crosstalk; Optical packet switching; Optical router; Optical switch fabric; Relative intensity noise; Transparent optical network; Wavelength routing.**

I. Introduction

Optical communications equipment deployed at the beginning of the millennium is now reaching its capacity limits [1]. Meanwhile, optical network traffic is booming due to web services, virtualization, and cloud computing. Content providers are migrating to large-scale computer warehouses [2,3] that offer high scalability and connectivity.

We address requirements for optically interconnected data centers (DCs) delivering high computing capacity [4,5]. Optical interconnection provides much greater bandwidth than electronic data center networks, enabling increased data rates and throughput [6]. It also offers a method to scale the port count to match the need of DCs with hundreds of thousands of interconnected nodes. It has been reported that data volumes and network bandwidth consumed by DCs double every 18 months. Meeting communication bandwidth demand is extremely challenging, driving research efforts to improve the capacity limits of optical fiber networks, for example, with the introduction of advanced modulation formats that increase spectral efficiency. The consolidation of infrastructure and resource virtualization requires reliable high-speed short-range interconnection with low power consumption and cost-effective solutions.

Scalability, i.e., the number of wavelengths per fiber, the switch size, and the resulting number of optical nodes that can be reached, is a very important issue for next-generation DCs that will incorporate highly integrated photonic modules. While the analysis in this paper uses bulk devices rather than a single integrated switch, adoption will be predicated on a cost-effective integrated solution in the long term—with the exception of optical delay lines and a limited number of erbium doped fiber amplifiers (EDFAs). Our focus is on determining what performance could be achieved, and whether this performance is adequate for DCs. Were arrayed waveguide gratings (AWGs) available with 72 input/output ports, and assuming 40 Gb/s operation, the proposed architectures could achieve multi-terabit-per-second switching capacities and service millions of cores with proper multiplexing.

Many electronic processing limitations can be overcome using an optical switch with AWGs. The silica-based planar lightwave circuit AWG has been widely used in wavelength-division multiplexing (WDM) systems. An $N \times N$ AWG can be realized by designing the free spectral range of the AWG to be N times the channel spacing [7,8]. The $N \times N$ AWG has a uniform-loss-cyclic-frequency property, as shown in Table I for $N = 8$, that can be exploited for wavelength-dependent routing.

Switch architectures combining tunable wavelength converters (TWCs) [9–15] and fiber delay lines (FDLs) provide tunable routing and contention resolution when used with AWGs, permitting ultrafast packet routing without optoelectronic conversion. An optical label containing routing information is added on the payload packet to allow all-optical signal processing regardless of the original payload format and bit rate. The switch scalability is affected by several physical layer limitations such as amplified spontaneous emission (ASE) noise and crosstalk. The TWC and optical amplifiers lead to accumulated ASE noise within the switch, including during recirculation in FDLs.

Manuscript received September 9, 2011; revised February 2, 2012; accepted April 9, 2012; published April 30, 2012 (Doc. ID 154413).

Q. Xu, Y. Ben M'Sallem, S. LaRochelle, and L. A. Rusch (e-mail: leslie.rusch@gel.ulaval.ca) are with the Center for Optics, Photonics and Lasers (COPL), ECE Department, Université Laval, Québec, QC G1V 0A6, Canada.

H. Rastegarfar and A. Leon-Garcia are with the Network Architecture and Services Group, ECE Department, University of Toronto, Toronto, Canada.

Digital Object Identifier 10.1364/JOCN.4.000412

TABLE I
8 × 8 AWG CONFIGURATION

		Output port							
		1	2	3	4	5	6	7	8
Input port	1	λ_1	λ_2	λ_3	λ_4	λ_5	λ_6	λ_7	λ_8
	2	λ_2	λ_3	λ_4	λ_5	λ_6	λ_7	λ_8	λ_1
	3	λ_3	λ_4	λ_5	λ_6	λ_7	λ_8	λ_1	λ_2
	4	λ_4	λ_5	λ_6	λ_7	λ_8	λ_1	λ_2	λ_3
	5	λ_5	λ_6	λ_7	λ_8	λ_1	λ_2	λ_3	λ_4
	6	λ_6	λ_7	λ_8	λ_1	λ_2	λ_3	λ_4	λ_5
	7	λ_7	λ_8	λ_1	λ_2	λ_3	λ_4	λ_5	λ_6
	8	λ_8	λ_1	λ_2	λ_3	λ_4	λ_5	λ_6	λ_7

Much current research focuses on the network layer in improving the optical packet switching (OPS) router throughput and minimizing the switching overheads. Little effort has focused on a practical system-level OPS interconnection solution to bridge the gap between the physical layer and the network layer. Gaudino *et al.* [16] gave a general framework to analyze the physical layer scalability and feasibility for N line cards, each carrying a single wavelength in a non-blocking architecture. To evaluate the signal-stage system performance without contention resolution and fiber delay line, they focused on the power sensitivity threshold at the receiver, the optical signal-to-noise ratio (OSNR) degradation due to the optical amplifier, and the AWG crosstalk level. In this paper, we delve into multi-stage architectures with multiple wavelengths per input port and contention resolution hardware. We also consider ASE in wavelength conversion, as well as amplification.

We consider four architectures that incorporate space, wavelength, and time for OPS router implementations. As mentioned in [17], in the integrated AWG module fabrication process, although the loss imbalance can be reduced to 1.3 dB, the frequency deviation scales with the square root of the port number, and the chip size increases in proportion to the port number, which limits the port number for the foreseeable future to under 100. Higher port number devices are also subject to lower fabrication tolerance and higher failure rate/yield. A more effective way to scale up is to use smaller port number components to build up a much larger dimension router [18–20].

Architecture A1 uses one wavelength per port in a single AWG plane, offering limited scalability due to the AWG size and TWC tuning range requirements. In order to extend the routing capacity, we propose the Clos-based [21,22] architectures A2, A3, and A4 that incorporate multi-stage designs when the routing needs to exceed the capacity of the largest feasible single AWG. Architectures A2 and A3 both have a four-stage configuration that consists of "line switching" and "wavelength switching", using multiple wavelengths per port at the third stage. The main differences between A2 and A3 are the position and the number of buffering FDLs. Architecture A4 is a three-stage design using only "line switching" at the input, and a single wavelength per port at the middle stage. A2, A3, and A4 use distributed switching structure.

Unlike previous research, we analyze the accumulated noise to quantify the signal Q-factor degradation in OPS routers [23]. As the total noise at the receiver is not captured by OSNR analysis (due to the AWG crosstalk), we calculate the independent noise contributions in a complete multi-stage system. We characterize the relevant optical components and their noise contributions. We also evaluate how recirculation of contending packets affects the overall performance for different architectures.

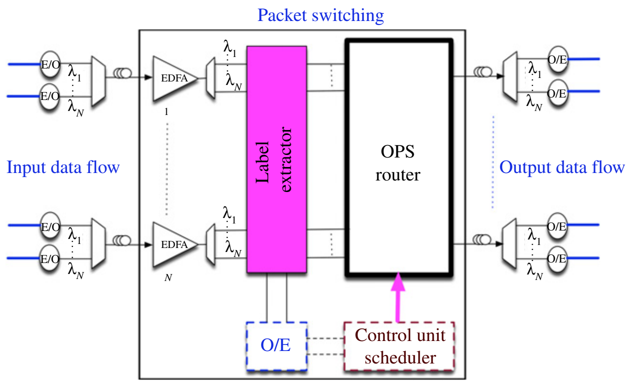


Fig. 1. (Color online) Non-blocking OPS router.

The paper is organized as follows. In Section II, we propose four non-blocking OPS router architectures for DC applications. In Section III, we describe analytical models of crosstalk and ASE noise for on–off-keying (OOK) signals. In Section IV, we compare architecture complexity via component count, as well as noise accumulation and Q-factor degradation, including the impact of packet recirculation. Finally, in Section V, we conclude and discuss next-generation OPS routers. We include three appendices that detail derivation of equations used for simulations in the body of the paper. Appendices A and B derive crosstalk expressions for the case of a single wavelength per port and multiple wavelengths per port, respectively. Appendix C quantifies the amplified spontaneous emission from EDFAs and wavelength conversion.

II. ARCHITECTURE OF OPS ROUTERS FOR DATA CENTERS

All OPS router architectures considered use AWGs for the wavelength-routed optical switch fabric. Contention is resolved using TWCs and FDLs. Input fibers carrying multiple wavelengths first have their signals demultiplexed into separate wavelength channels. As shown in Fig. 1, each of these demultiplexed channels is input to a label extraction module that accesses routing information and converts it to electronic form. A centralized switch control unit exploits this information to configure TWCs within the switch using an appropriate algorithm. Contention is identified and resolved to determine switch reconfiguration and buffering.

For simplicity, we assume N input ports, with the same number, N, of wavelengths per port. Our analysis could easily be extended to the more general case of M wavelengths on each of the N ports. We also assume equal priority across ports, and furthermore that the data traffic is pre-balanced. This assumption of load-balanced uniform traffic precludes analysis of traffic patterns that is outside the scope of this paper, and justifies analysis of average crosstalk as opposed to the highly unlikely worst-case crosstalk.

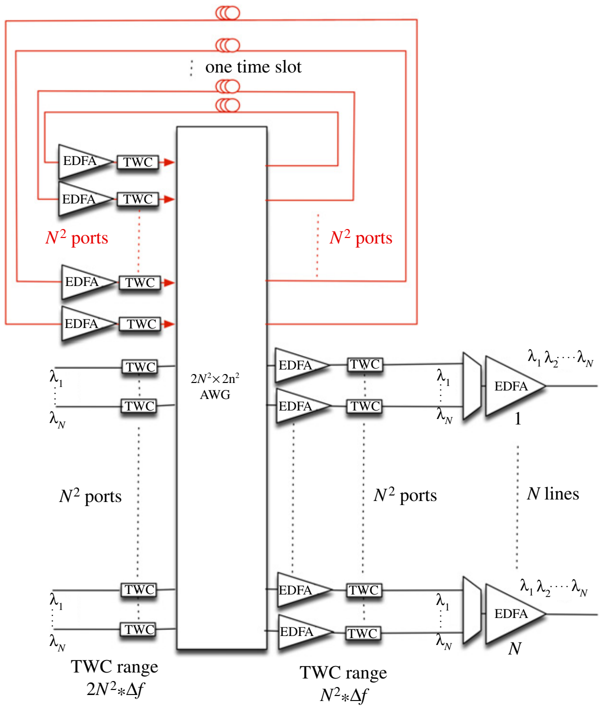


Fig. 2. (Color online) Single-stage OPS router architecture (A1).

This router scheme offers network management flexibility, as any incoming wavelength at any port can be routed to a designated port with a desired wavelength, i.e., non-blocking [6,24]. In this section, we propose four non-blocking OPS router architectures that incorporate space, wavelength, and time switching. We focus on the OPS router (heavy border inside the packet switching block), as the label extractor unit is the same for all architectures.

A. Architecture A1: Single-stage OPS Router

Figure 2 depicts the single-stage architecture A1 [25] that consists of a $2N^2\times 2N^2$ AWG, N^2 FDLs, and $3N^2$ TWCs. The $2N^2$ TWCs at the input of the AWG are used to select the intended output port for the fresh incoming packets, as well as the previously buffered packets. The N^2 TWCs at the output of the AWG convert the wavelengths of outgoing optical packets to the appropriate transmission wavelengths before leaving and being multiplexed onto the output switch port.

A major drawback of architecture A1 is the technological limitations on the number of AWG ports. The number of required AWG ports scales as the product of the number of input fibers and the number of input wavelengths per fiber; in the present work, this is N^2. Large port number also requires stringent tuning range requirements on the TWCs. For a $2N^2\times 2N^2$ AWG, the TWC tuning range must cover $2N^2*\Delta f$, where Δf is the frequency spacing of the AWG. As will be discussed in detail in Section III, the main advantage of this architecture is its low intraband crosstalk and simple design.

B. Architecture A2: Four-stage OPS Router

We propose a multi-stage architecture to overcome the scalability limitations of the single-stage architecture A1. Our architecture consists of four stages, as depicted in Fig. 3. The proposed architecture is rearrangeably non-blocking, featuring a Clos-based design, with an equal number of switches (planes) per stage. In this design, the third-stage distribution AWG sets up single paths between any pair of stage 2 and stage 4 planes, thus implementing the shuffle of wires between the last two stages of the Clos network.

The first stage in A2 consists of N planes, each performing non-blocking switching among the packets belonging to the same input fiber. Each first-stage plane comprises a $2N\times 2N$ AWG, $2N$ TWCs, and N FDLs, and combines two functions:

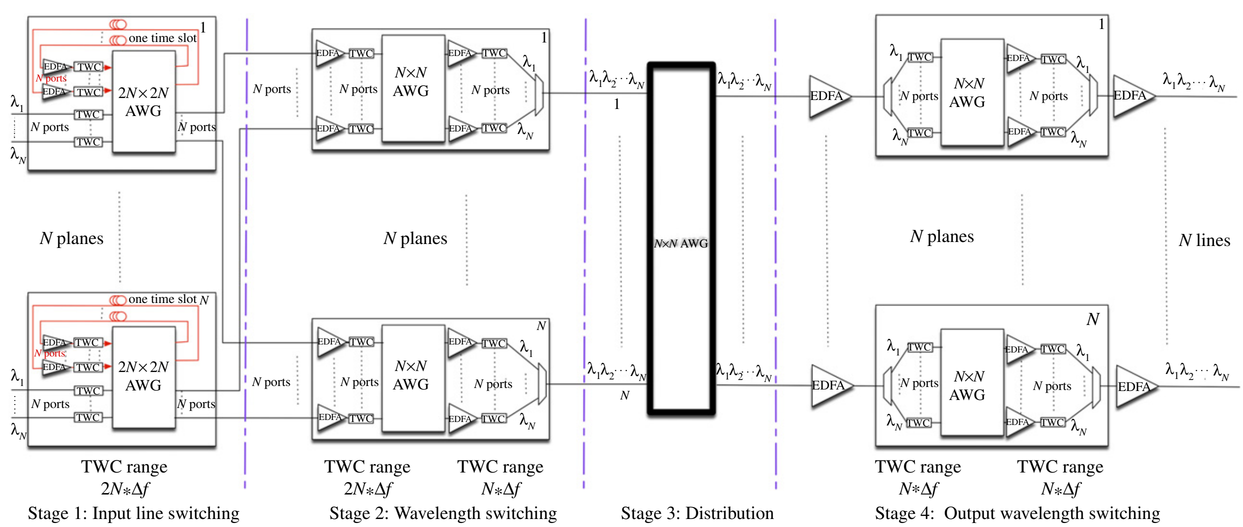


Fig. 3. (Color online) N-plane four-stage OPS router (A2).

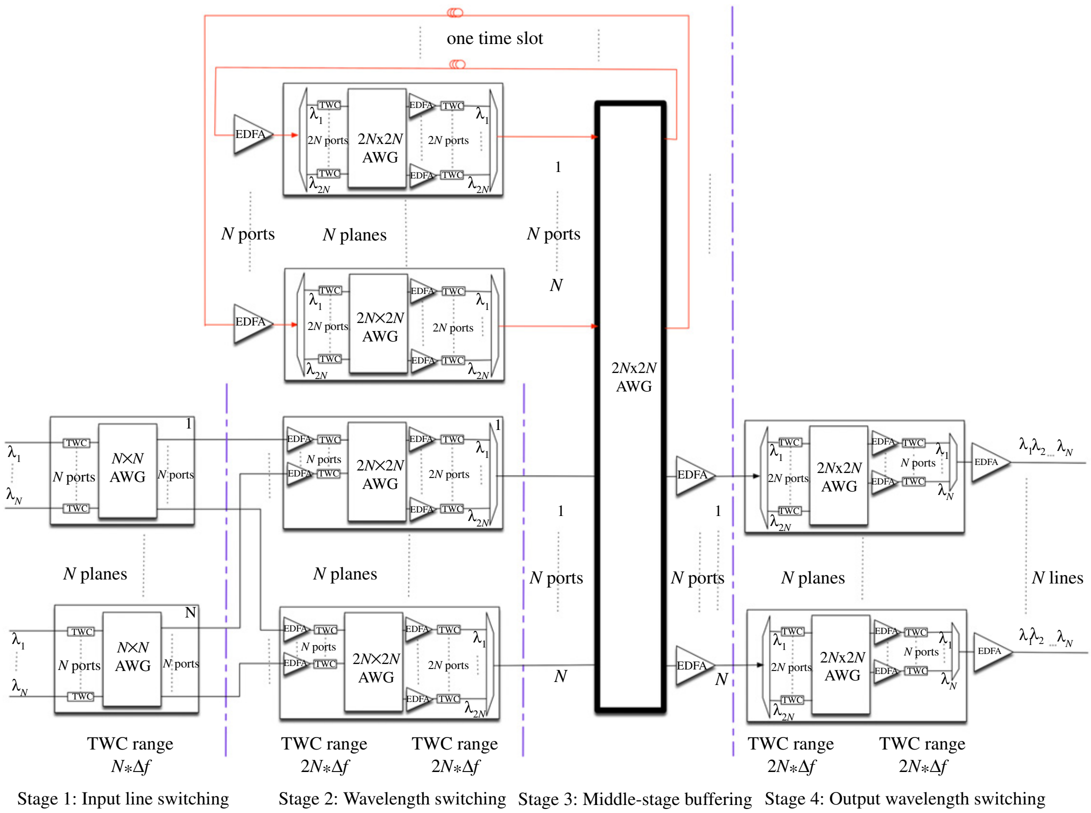


Fig. 4. (Color online) N-plane middle-stage buffering OPS router (A3).

selection of packets that proceed forward into the second stage of the Clos network (corresponding to the lower half of the $2N \times 2N$ AWG) and recirculation of contending packets (corresponding to the upper half of the $2N \times 2N$ AWG). The output packets from each plane are distributed among N second-stage planes. Each second-stage plane consists of an $N \times N$ AWG and $2N$ TWCs, allowing the rearrangement of the wavelengths for routing purposes before forwarding the packets to the third stage. The third stage consists of an $N \times N$ AWG that forwards the packets to their designated output fiber. Finally, in the fourth stage, each of the N planes consists of an $N \times N$ AWG and $2N$ TWCs, and allows for the rearrangement of the wavelengths such that each packet is switched to the designated output wavelength on the appropriate output port.

The main advantage of A2 is its scalability, as a high port count optical switch fabric can be realized using low port count AWGs [26]. The total number of input AWG ports in A2 is $4N^2+N$, but no single AWG has more than $2N$ input ports. The number of TWCs has climbed from $3N^2$ to $6N^2$; however, the TWC tuning range requirement has been drastically reduced from $2N^2 * \Delta f$ to $2N * \Delta f$.

C. Architecture A3: Middle-stage Buffering OPS Router

Architecture A2 requires more components than A1 in the router but offers much higher scalability. One common drawback of A1 and A2 is the high number of FDLs required for providing a certain number of buffer spots. We therefore propose an alternative approach using the “N-plane middle-stage buffering” architecture A3, shown in Fig. 4. The first and second stages are similar to those of A2; in stage 1, each plane consists of an $N \times N$ AWG and N TWCs; and in stage 2 each plane consists of a $2N \times 2N$ AWG and $3N$ TWCs.

Instead of buffering the contending packets in the input switching stage, in A3, N FDLs are implemented in the third stage, each carrying multiple wavelengths and connected to a buffer plane that consists of a $2N \times 2N$ AWG and $4N$ TWCs. The buffered packets are again demultiplexed at the input to the buffer plane, sending one wavelength to an AWG input port. Each buffered packet is switched to the designated output wavelength and reenters the middle-stage $2N \times 2N$ AWG after recirculation.

The main advantage of this architecture compared to A2 is that the number of FDLs is drastically reduced from N^2 to N, while still offering improved scheduling flexibility and contention resolution.

D. Architecture A4: Clos Network OPS Router

Architectures A2 and A3 both require a single-plane distribution stage that suffers from intense crosstalk in the AWG. A distributed fabric using an N-plane distribution stage (Clos network), Fig. 5, can mitigate intense AWG crosstalk [21,22].

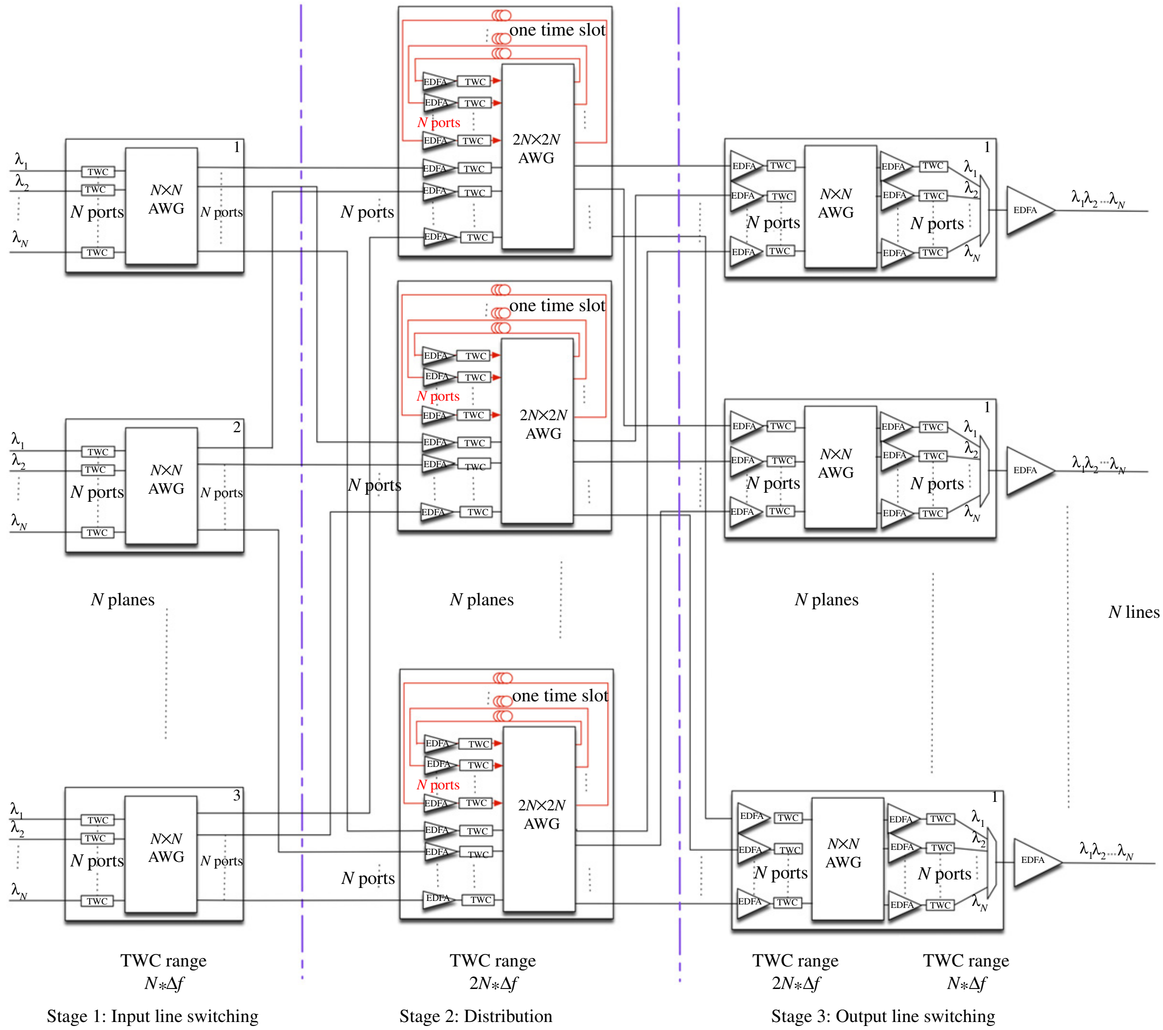


Fig. 5. (Color online) Clos network OPS router (A4).

The first stage consists of N planes, each performing non-blocking switching among the packets belonging to the same input fiber. Each plane is comprised of N TWCs and an $N \times N$ AWG. The output packets from each plane are distributed among N second-stage planes. Each plane routes contending ("losing") packets to the FDL buffers, and forwards the "winning" packets to appropriate output ports. Each second-stage plane consists of a $2N \times 2N$ AWG, $2N$ TWCs, and N FDLs. The third stage consists of N planes, and it allows for rearranging the wavelength. Each plane consists of an $N \times N$ AWG and $2N$ TWCs. Each packet is switched to the designated output wavelength and leaves the switch on the appropriate output port.

III. SNR Degradation

The characterization of the optical devices used in the architectures described previously is crucial to effectively assess their scalability and switching performance. This section presents analysis of SNR degradation. The crosstalk induced in the passive devices and the ASE introduced by active devices within the optical switch architectures are quantified in the appendices. As data center interconnection involves short-distance transmission in a "warehouse" [2], in this analysis we do not include fiber attenuation, chromatic dispersion, polarization mode dispersion, and high power-induced fiber nonlinear effects.

For a binary modulation format, the Q-factor of OOK signals is given by

$$Q = \frac{(I_1 - I_0)}{\sigma_{\text{mark}} + \sigma_{\text{space}}}, \tag{1}$$

where $I_0 = 0$ and $I_1 = E^2$ are the average photocurrent of space bit "0" and mark bit "1", and $\sigma_{\text{space}}, \sigma_{\text{mark}}$ are the noise level for space and mark, respectively.

Accordingly, the bit error rate (BER) can be given by

$$\text{BER} = \frac{1}{2}\text{erfc}\left(\frac{Q}{\sqrt{2}}\right) \approx \frac{\exp\left(-Q^2/2\right)}{Q\sqrt{2\pi}}. \tag{2}$$

We assume an input power of 2 dBm to ensure efficient wavelength conversion [9,14]. For the sake of simplicity, for each architecture, we assume that EDFAs compensate losses so that the signal power is maintained fixed at each TWC input (to ensure good conversion efficiency). This balance applies to paths with and without recirculation. We assume that each input wavelength port of the switch is occupied with probability L of a signal being present, i.e., wavelength load.

A. Noise Contributions

In Appendix A, we examine the level of crosstalk when an AWG has one wavelength per input port. We find the crosstalk noise to be

$$\begin{aligned}\sigma^2_{\mathrm{AWG}-1\lambda} &= X(N,\rho)|_{\rho=L/N} \\ &= E^4 R_{NX} L \frac{N-1}{N} + E^4 (R_{AX} - R_{NX}) \\ &\quad \times \sum_{i=1}^{N-1} \mathrm{PMF}(i)[P_X(1,i) + 2P_X(2,i)], \end{aligned} \tag{3}$$

where R_{AX} is the adjacent and R_{NX} the non-adjacent crosstalk power ratio, and PMF(i) and $P_X(\cdot,i)$ are the crosstalk probabilities defined in Appendix A.

In Appendix B, we examine the level of crosstalk when an AWG has multiple wavelengths present at each input port. We find the crosstalk noise to be

$$\sigma^2_{\mathrm{AWG}-m\lambda} = X(N,\rho)|_{\rho=L} \tag{4}$$

for a mark; the crosstalk is zero for a space. The crosstalk introduced by demultiplexing these signals at the receiver (see Appendix A) is

$$\begin{aligned}\sigma^2_D &= X_D(N,\rho)|_{\rho=L} \\ &= E^4 R_{D-NX}\rho(N-1) + E^4(R_{D-AX} - R_{D-NX}) \\ &\quad \times \sum_{i=1}^{N-1} \mathrm{PMF}(i)[P_x(1,i) + 2P_x(2,i)], \end{aligned} \tag{5}$$

where R_{D-AX} is the adjacent and R_{D-NX} the non-adjacent crosstalk power ratio for the demultiplexer.

In Appendix C, we find the ASE contributions for the EDFA and TWCs to be

$$\begin{cases} \sigma^2_{\mathrm{ASE-EDFA}} = -31\ \mathrm{dBm} \cdot \dfrac{1.4R_S}{12.5\ \mathrm{GHz}} \\ \sigma^2_{\mathrm{ASE-TWC}} = -36\ \mathrm{dBm} \cdot \dfrac{1.4R_S}{12.5\ \mathrm{GHz}}. \end{cases} \tag{6}$$

In the following sections, we address each architecture in turn and find the noise contributions from the various stages of EDFA, TWC, single and multiple wavelength input AWGs, and demultiplexer, as appropriate. We will allow for the possibility of B recirculations to investigate each architecture's scalability and routing performance. We assume that the numbers of input ports and buffer ports are both equal to N.

B. Balanced Chain

To simplify analysis, we make very general assumptions on the gain in amplification and losses for all devices. We refer to this as the balanced chain. We assume that the input signal power to any TWC is fixed—any losses incurred (recirculation or direct) have been precisely balanced by a gain in amplification to maintain the signal power at these inputs. For the architectures examined, this assumption is consistent with the current state of device technology. The TWC signal input power is assumed to be 2 dBm to obtain good conversion efficiency, assumed to be −10 dB [9,14,15]. The loss in wavelength conversion and insertion losses (typically 6 dB for an AWG and 2 dB for integrated mux/demux) are compensated by EDFA gain on the order of 18 dB. The EDFA input power will be −16 to −14 dBm with the numbers quoted; we assume that the EDFA gain is tweaked to precisely balance the chain.

The total ASE contribution (signal-spontaneous beating) for a chain of m EDFAs and n TWC stages is

$$\sigma^2_{ASE-chain} = m \cdot \sigma^2_{\mathrm{ASE-EDFA}} + n \cdot \sigma^2_{\mathrm{ASE-TWC}}. \tag{7}$$

The EDFA stage before the label extractor, shown in Fig. 1, is not counted, as we only calculate the noise accumulation inside the OPS router.

C. Noise of OPS Router A1

For A1 in the $2N^2 \times 2N^2$ AWG, the single-stage crosstalk noise is due to the one wavelength per input port scenario:

$$\sigma^2_{(A1)-1} = \sigma^2_{ASE-1\lambda} = X\left(2N^2,\rho\right)|_{\rho=L/2N^2}. \tag{8}$$

Taking into account B recirculations, for a space bit, the only noise is the demux interband crosstalk:

$$\sigma^2_{space-A1} = X_D(N,\rho)|_{\rho=L}. \tag{9}$$

For a mark, we accumulate the crosstalk and ASE once for each recirculation, so the total noise is

$$\begin{aligned}\sigma^2_{\mathrm{mark}-A1} &= (1+B)\sigma^2_{(A1)-1} + X_D\left(2N^2,\rho\right)|_{\rho=L} \\ &\quad + (2+B)\sigma^2_{\mathrm{ASE-EDFA}} + (2+B)\sigma^2_{\mathrm{ASE-TWC}}. \end{aligned} \tag{10}$$

D. Noise of OPS Router A2

For A2, in the first, second, and fourth stages, the AWGs accumulate crosstalk based on a one wavelength per port scenario, while in the third stage the AWG accumulates crosstalk in a multiple wavelength per port scenario.

$$\sigma^2_{(A2)-1} = X(2N,\rho)|_{\rho=L/2N} \tag{11}$$

$$\sigma^2_{(A2)-2} = X(N,\rho)|_{\rho=L/N} \tag{12}$$

$$\sigma^2_{(A2)-3} = X(N,\rho)|_{\rho=L} \tag{13}$$

$$\sigma^2_{(A2)-4} = X(N,\rho)|_{\rho=L/N}. \tag{14}$$

Taking into account B recirculations, for a space bit,

$$\sigma^2_{space-A2} = X_D(N,\rho)|_{\rho=L}. \tag{15}$$

For a mark,

$$\begin{aligned}\sigma^2_{\text{mark}-A2} &= (1+B)\sigma^2_{(A2)-1}+\sigma^2_{(A2)-2}+\sigma^2_{(A2)-3}+\sigma^2_{(A2)-4}\\ &+X_D(2N^2,\rho)|_{\rho=L}+(5+B)\sigma^2_{\text{ASE-EDFA}}\\ &+(5+B)\sigma^2_{\text{ASE-TWC}}.\end{aligned} \tag{16}$$

E. Noise of OPS Router A3

A3 is similar to A2, except that the recirculation shared buffer lines are between the second and the third stages:

$$\sigma^2_{(A3)-1} = X\left(N,\rho\right)|_{\rho=L/N}. \tag{17}$$

In the second-stage $2N\times 2N$ AWG, the lower N planes only have N input channels each (direct path), while the upper N planes can have $2N$ input channels each (recirculation) at a given time slot:

$$\sigma^2_{(A3)-2} = \begin{cases}\sigma^2_{(A3)-2D} = X\left(2N,\rho\right)|_{\rho=0.5L/2N} & \text{direct path}\\ \sigma^2_{(A3)-2R} = X\left(2N,p\right)|_{\rho=L/2N} & \text{recirculation}\end{cases} \tag{18}$$

$$\sigma^2_{(A3)-3} = X\left(2N,\rho\right)|_{\rho=L} \tag{19}$$

$$\sigma^2_{(A3)-4} = X\left(2N,\rho\right)|_{\rho=0.5L/2N}. \tag{20}$$

Taking into account B recirculations, for a space bit,

$$\sigma^2_{space-A3} = X_D\left(N,\rho\right)|_{\rho=L}. \tag{21}$$

For a mark,

$$\begin{aligned}\sigma^2_{\text{mark}-A3} &= \sigma^2_{(A3)-1}+\left(\sigma^2_{(A3)-2D}+B\cdot\sigma^2_{(A3)-2R}\right)\\ &+(1+B)\cdot\sigma^2_{(A3)-3}+\sigma^2_{(A3)-4}+X_D\left(N,\rho\right)|_{\rho=L}\\ &+(5+2B)\sigma^2_{\text{ASE-EDFA}}\\ &+(5+2B)\sigma^2_{\text{ASE-TWC}}.\end{aligned} \tag{22}$$

F. Noise of OPS Router A4

In architectures A2 and A3, the third stage for distribution requires a single AWG, thus introducing severe intraband crosstalk. The Clos architecture can overcome this impairment by using an N-plane distribution stage in which data only suffers crosstalk under the single wavelength per port scenario; hence

$$\sigma^2_{(A4)-1} = X\left(N,\rho\right)|_{\rho=L/N} \tag{23}$$

$$\sigma^2_{(A4)-2} = X\left(2N,\rho\right)|_{\rho=L/2N} \tag{24}$$

$$\sigma^2_{(A4)-3} = X\left(N,\rho\right)|_{\rho=L/N}. \tag{25}$$

Taking into account B recirculations, for a space bit,

$$\sigma^2_{space-A4} = X_D\left(N,\rho\right)|_{\rho=L}. \tag{26}$$

For a mark,

$$\begin{aligned}\sigma^2_{\text{mark}-A4} &= \sigma^2_{(A4)-1}+(1+B)\sigma^2_{(A4)-2}\\ &+\sigma^2_{(A4)-3}+X_D\left(N,\rho\right)|_{\rho=L}\\ &+\left((4+B)\sigma^2_{\text{ASE-EDFA}}+(4+B)\sigma^2_{\text{ASE-TWC}}\right).\end{aligned} \tag{27}$$

IV. OPS Router Performance Analysis

In this section, we investigate the OPS architecture scalability and recirculation performance due to the noise accumulation in the multi-stage router systems.

A. Devices in OPS Routers

We first analyze the scalability limitations with respect to the port count of the proposed architectures. In order to increase the port count of such switch fabrics, one direct method is to increase the AWG size. However, there are technological limits to the current AWG size, such as the fabrication-induced phase error that increases with AWG size. In [17], Kamei *et al.* estimate that the AWG frequency deviation increases in proportion to the square root of the AWG size. Moreover, the on-chip photonics integration design and the fabrication process impose scaling limitations, and the TWC effective tuning range also imposes an upper limit to the AWG size to be used in the switch fabric. In the following analysis, we consider a maximum AWG size of 72×72. Thus architecture A1 can support at most 36 input channels, while the other three architectures can support 1296.

We show in Table II the required tuning range and the number of input AWG ports, FDLs, TWCs, and EDFAs, optical interconnections, and number of wavelengths in the router for a given maximum number of input channels (N^2) and input fiber links (N). Architecture A1 has a scale limit because the AWG size is $2N^2\times 2N^2$, and hence $N_{\max}=6$, whereas, for architectures A2, A3, and A4, the required AWG size is $2N$, and hence $N_{\max}=36$. In A3, the buffer sharing at the third stage reduces the number of FDLs from N^2 to N. We also notice in Table II that, compared to A1, the number of wavelength in A2, A3, and A4 has been reduced from $2N^2$ to $2N$, which relaxes the requirements on the TWC tuning range. Please note that our analysis assumes uniform TWC conversion efficiency across the tuning range. Because of its lack of scalability, we will not discuss architecture A1 further in this paper.

B. Noise Contributions of OOK Payloads

There are three main AWG characteristics that directly impact the OPS router performance: the loss imbalance, the deviation from the WDM grid center frequency, and the crosstalk power ratio. The standard planar lightwave circuit AWGs usually have a relatively high crosstalk ratio. Kamei *et al.* [17] reported integrated AWG modules that reduce

TABLE II
DEVICE REQUIREMENTS IN OPS ARCHITECTURE WITH PORT COUNT N

	A1	A2	A3	A4
Number of input AWG ports	$2N^2$	$4N^2+N$	$4N^2+2N$	$4N^2$
Maximum AWG size	$2N^2 \times 2N^2$	$2N \times 2N$	$2N \times 2N$	$2N \times 2N$
Number of FDLs	N^2	N^2	N	N^2
Number of TWCs	$3N^2$	$6N^2$	$11N^2$	$5N^2$
Number of EDFAs	$2N^2+N$	$4N^2+2N$	$6N^2+3N$	$5N^2$
Optical interconnections	0	N^2+2N	N^2+3N	$2N^2$
TWC tuning range	$2N^2 * \Delta f$	$2N * \Delta f$	$2N * \Delta f$	$2N * \Delta f$
Number of wavelengths	$2N^2$	$2N$	$2N$	$2N$

these three key parameters considerably; however, the port number of AWG modules remains limited due to the available bandwidth range and the fabrication error tolerance.

In this section, we consider 1) commercial AWGs and 2) low-crosstalk AWGs from research labs. We use typical device parameter values based on manufacturer data sheets. For the Demux, $R_{D-NX} = -50$ dB, $R_{D-AX} = -25$ dB [27]. For the commercial AWG, $R_{\mathrm{AWG}-NX} = -30$ dB, $R_{\mathrm{AWG}-AX} = -25$ dB [28]. For the low-crosstalk AWG router module, $R_{\mathrm{AWG}-NX} = -43$ dB, $R_{\mathrm{AWG}-AX} = -31$ dB [17].

For a given NRZ OOK signal symbol rate R_S, we consider electrical receiver bandwidth $\Delta f = 0.7R_S$. We also assume the optical filter equivalent bandwidth $\Delta v_r \approx 2\Delta f = 2 * 0.7R_S$. We examine two symbol rates: $R_S = 10$ Gb/s and $R_S = 40$ Gb/s.

In Section III, we developed the noise variance for marks and spaces for each architecture. These variances are used in Eqs. (1) and (2) to assess the BER. Contributions from various components (EDFA, TWC, AWG crosstalk) are isolated to see their individual contribution to the noise power. We assume that the router is fully loaded with $L = 1$, and we examine the noise contributions for packets with no recirculation ($B = 0$) and with 10 recirculations ($B = 10$). Noise power contributions are presented in Fig. 6 as a function of switch size (number of ports).

When the AWG port count is small, adjacent channels induce severe crosstalk even in a single wavelength per port scenario. We notice in Fig. 6 that the crosstalk-induced noise first reaches a peak at $N = 3$, mainly due to the severe adjacent-channel crosstalk. As A2 and A3 both include the multiple wavelength per port scenario, the crosstalk-induced noise increases gradually with N; the TWC ASE noise is higher than the EDFA ASE noise, as expected.

For $R_S = 10$ Gb/s, crosstalk is the main contributor to the signal degradation in A2 and A3 with a standard AWG, because of the intense AWG crosstalk. However, if we replace the standard AWG by a low-crosstalk AWG, the crosstalk-induced noise and the ASE-induced noise are both dominant impairments to the signal. In A4, the AWG-induced crosstalk is all due to the single wavelength per port scenario. If we consider only a large port count AWG (e.g., $N > 10$), the probability of having adjacent-channel crosstalk, as well as the crosstalk noise, decreases gradually with N. The AWG-induced noise is dominant when using the standard AWG, and the TWC ASE becomes dominant when using the low-crosstalk AWG. The above conclusions are valid for both $B = 0$ and 10, and the low-crosstalk AWG considerably alleviates the overall noise impairment, especially as N scales up.

For $R_S = 40$ Gb/s, the ASE-induced noise becomes much higher due to the filter and receiver bandwidth. In A2, the TWC ASE noise contribution becomes comparable to the crosstalk-induced noise using the standard AWG, especially for packets going through many recirculations. In A3, the standard AWG crosstalk-induced noise is still considerably higher than the other noise contributions, while using the low-crosstalk AWG can reduce the total noise in spite of the high TWC ASE noise. Finally, in A4, as no intense crosstalk is present, the TWC ASE noise is dominant over the AWG crosstalk, and hence becomes the main impairment.

Regarding recirculations, the noise accumulation in A3 is the most severe, since the packets in each recirculation loop will suffer from multiple wavelength per port AWG crosstalk, requiring more TWCs and EDFAs. Hence the noise increases rapidly, as we can see in Fig. 6, although the low-crosstalk AWG modules can significantly reduce the crosstalk noise, by a factor of ~10.

We recall that the above crosstalk analysis is conducted by assuming all input signals to have the same polarization, i.e., a worst-case assumption. For random polarization states, the crosstalk-induced noise can be reduced by one half [23].

C. Q-factor Degradation and Packet Recirculation

We now examine Q-factor degradation. We consider two typical traffic loads: $L = 0.5$ as a standard router load and $L = 1$ as an “outage” load value. In Fig. 7, we show the Q-factor degradation with regards to the recirculation number B, for $N = 36$, corresponding to the large-scale routers. We assume that a Q-factor $Q \approx 6$ is required for bit error rate BER $= 10^{-9}$.

In A2, the packets go through intense AWG crosstalk only once and suffer only slight crosstalk in the recirculation. While there is no intense AWG crosstalk in packet recirculation, the crosstalk-induced noise in the single wavelength per port AWG scenario is comparable or higher than the TWC and EDFA ASE noise. Therefore, using the low-crosstalk AWG will considerably improve the routing performance. At 10 Gb/s, the Q-factor remains above 6 using the low-crosstalk AWG after 27 and 24 recirculations for $L = 0.5$ and 1.0, respectively. At 40 Gb/s, due to the higher ASE power, the packets can only recirculate four times even with a low-crosstalk AWG.

Buffer line sharing in A3 is not a good choice, as each recirculation will include additional TWC and EDFA ASE noise, as well as intense crosstalk. Even with a low-crosstalk AWG, for a 10 Gb/s signal, the Q-factor falls below 6 after

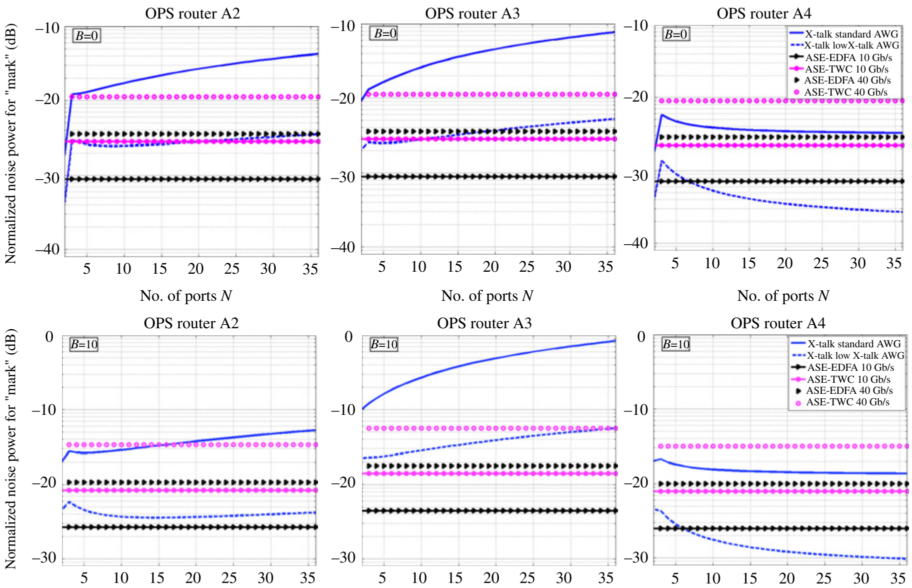


Fig. 6. (Color online) Noise contributions from crosstalk, EDFA, and TWC using a standard AWG and a low-crosstalk AWG. $L = 1, B = 0$ (upper), $B = 10$ (lower).

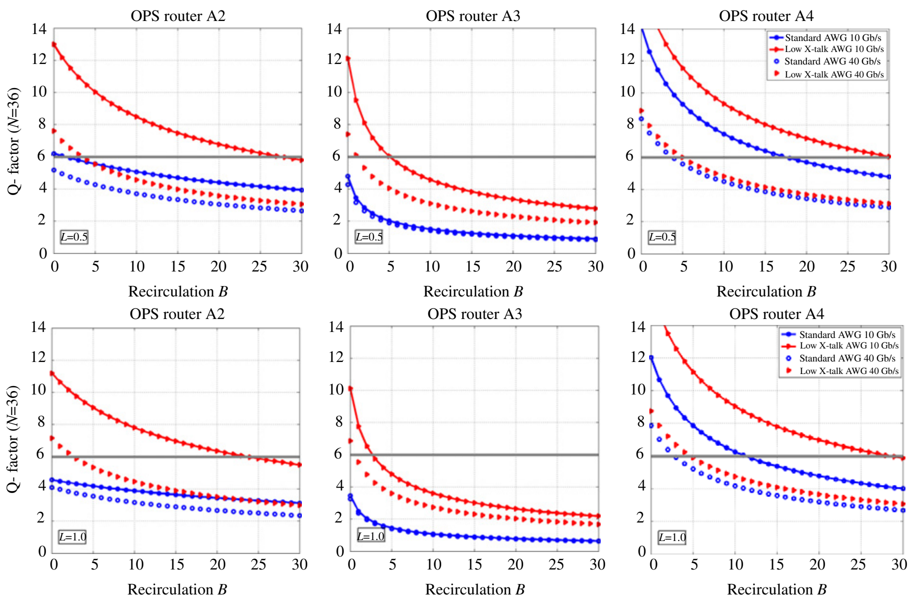


Fig. 7. (Color online) Evaluation of Q-factor with recirculation times B for $N = 36, L = 0.5$ (upper), $L = 1.0$ (lower). The thick solid line is $Q = 6$ (error free).

five recirculations for $L = 0.5$, and after three recirculations for $L = 1.0$, while, for a 40 Gb/s signal, the packets can only be recirculated a couple of times.

Compared to A2, in A4 the packets only go through slight crosstalk. At 10 Gb/s, using a standard AWG, packets can undergo 18 and 12 recirculations for $L = 0.5$ and 1.0, respectively; a low-crosstalk AWG can increase the recirculation times to around 30. At 40 Gb/s, as seen in Fig. 6, the TWC ASE becomes the main noise contribution; hence using a low-crosstalk AWG cannot notably increase the packet recirculation times. The large-scale system implementation in A4 requires more optical interconnections due to the second-stage distributed multi-plane architecture, but it exhibits better tolerance to signal degradation caused by packet buffering.

AWG crosstalk and TWC ASE noise are the main impairments to OPS routers in our studies. Additional ASE accumulation is inevitable, but intense crosstalk can be avoided by using architectures A2 and A4. In moving from 10 Gb/s to 40 Gb/s, the TWC conversion efficiency may degrade, and the signal OSNR decline. Advanced modulation formats can improve the bit rates and routing performance, but meanwhile increase the receiver complexity and cost; therefore a trade-off must be found. Finally, OPS architectures should be designed compatible with photonics on-chip integration, i.e., offering compact, low power consumption, and high-performance solutions.

V. Conclusion

We have proposed four non-blocking OPS router architectures including contention resolution, for data center applications, in which A1 includes a single-stage design and A2, A3, and A4 incorporate multi-stage designs when the routing needs exceed the capacity of the largest feasible single AWG. We have developed an analytical framework tracking noise accumulation in multi-stage OPS routers to analyze OOK signal impairments. The model considers AWG crosstalk under single and multiple wavelengths per input port scenarios, Demux crosstalk, and ASE from TWCs based on semi-conductor optical amplifiers and EDFAs.

We examined attainable switch sizes given trends in AWG technology, as well as an analysis of signal degradation (Q-factor) as a function of recirculations, for each of the three multi-stage architectures. We showed that TWC ASE noise and AWG intraband crosstalk are the dominant impairments, especially the high-level AWG crosstalk when a multiple wavelengths per input port scenario applies.

Based on our routing analysis, architecture A4, which uses distributed N-plane fabric, suffers from moderate crosstalk and is our most promising design in terms of scalability and signal quality. Recently developed low-crosstalk AWGs would further improve the routing performance and enable large-scale all-optical router implementation.

Appendix A: AWG Crosstalk Analysis for Single Wavelength per Port

Consider an $N\times N$ AWG loaded with a single wavelength per port. We assume that an input port is occupied with probability $L\le 1$, i.e., the load. We assume that one of the N wavelengths appears with uniform probability at each port. Furthermore, we assume that the packet destinations are uniformly distributed across the output ports. For a given output port, the crosstalk seen by any given wavelength can come from any of $N-1$ input ports, each with probability $\rho = L/N$. Let i be the number of input signals on the same wavelength contributing to the crosstalk of an arbitrary desired signal, i.e., the intraband crosstalk terms. The probability mass function of i is binomial,

$$\mathrm{PMF}(i)=\binom{N-1}{i}\rho^{i}\cdot(1-\rho)^{N-1-i}, \tag{28}$$

with mean $L(N-1)/N$.

The intraband crosstalk contribution from physically adjacent ports is much greater than that of other ports. Most component manufacturers characterize performance in terms of adjacent (R_{AX}) and non-adjacent (R_{NX}) crosstalk power ratios. Therefore, we examine the probability of i_{AX}, the number of adjacent ports occupied by the same wavelength as the desired signal, $i_{AX}\in\{0,1,2\}$. Let $P_X(i_{AX},i)$ be the probability of having i_{AX} adjacent crosstalk terms when there are i crosstalk terms,

$$\begin{aligned}
P_X(i_{AX}=0,\ i)&=\frac{\binom{2}{0}\binom{N-3}{i}}{\binom{N-1}{i}} \quad 1\le i\le N-3\\
P_X(i_{AX}=1,\ i)&=\frac{\binom{2}{1}\binom{N-3}{i-1}}{\binom{N-1}{i}} \quad 1\le i\le N-2\\
P_X(i_{AX}=2,\ i)&=\frac{\binom{2}{2}\binom{N-3}{i-2}}{\binom{N-1}{i}} \quad 2\le i\le N-1.
\end{aligned} \tag{29}$$

Let $X(\rho,N)$ be the expected value of the noise contribution from the intraband crosstalk for a single wavelength per port AWG of size $N\times N$, and effective offered load $\rho = L/N$. Then

$$\begin{aligned}
X(N,\rho) &= E[X]=E^4R_{AX}E[i_{AX}]+E^4R_{NX}E[i-i_{AX}]\\
&=E^4R_{NX}\rho(N-1)+E^4(R_{AX}-R_{NX})E[i_{AX}]\\
&=E^4R_{NX}\rho(N-1)+E^4(R_{AX}-R_{NX})E[E[i_{AX}|i]]\\
&=E^4R_{NX}\rho(N-1)+E^4(R_{AX}-R_{NX})\sum_{i=1}^{N-1}\mathrm{PMF}(i)\\
&\quad\times[P_X(1,i)+2P_X(2,i)],
\end{aligned} \tag{30}$$

where E is the optical signal field.

Appendix B: Crosstalk for Multiple Wavelengths per Port

To develop expressions for the crosstalk in architectures with AWGs with multiple wavelengths per port, we examine a representative model in Fig. 8. As crosstalk ultimately develops in the photodetector, we focus on the crosstalk following demultiplexing of the output WDM signal and optoelectrical conversion. While for a single wavelength per port we calculated the crosstalk based on statistics of channel occupancy (Appendix A), for multiple wavelengths per channel, we adopt a simpler analysis. In this appendix, we first calculate the crosstalk for full load ($L=1$). In Section IV we apply lower L, as we assume either all wavelengths are present or none, with probability L.

Let $E_{i,,j}$ be the optical field due to the signal on λ_i at switch input port j. Due to the cyclic nature of the AWG passive routing, the signals at output port p of the AWG without any crosstalk would be

$$\begin{aligned}
&\{E_{1,[N+1-p]_N+1}E_{2,[N+2-p]_N+1}\cdots\\
&E_{N-1,[2N-1-p]_N+1}E_{N,[2N-p]_N+1}\}\\
&=\left\{E_{i,[N+i-p]_N+1}\right\}_{i=1}^{N}\equiv\{E_{i,f_i(p)}\}_{i=1}^{N},
\end{aligned} \tag{31}$$

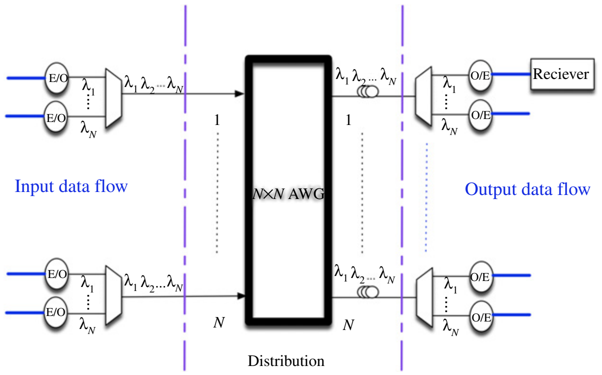


Fig. 8. (Color online) Representative architecture model.

where $[]_N$ is modulo N summation and $fi(p)$ is the function indicating the originating port of the signal on λ_i that is routed to output port p.

In addition to these desired signals, interference is present due to crosstalk at the AWG and demux outputs. For the AWG, each wavelength will see $N-1$ other input signals contributing to *intraband* crosstalk. It will also see *interband* crosstalk from N^2-N other signals on wavelengths other than λ_i. An optical demux or AWG is essentially a band-pass filter for each channel, which significantly reduces the interband crosstalk. In a WDM system, the demux prior to reception always isolates one wavelength; hence AWG interband crosstalk has much less impact than AWG intraband crosstalk [23,29]. We neglect the interband crosstalk from an AWG for these reasons. The demux has N signals on a single input port; hence the output ports see crosstalk, in this case strictly interband crosstalk. This interband crosstalk is not mitigated by additional optical filtering, and therefore cannot be neglected.

The output at port p on wavelength i including AWG crosstalk can be given by

$$\underbrace{E_{i,f_i(p)}}_{\text{signal}}+\underbrace{\sum_{\substack{k=1\\k\neq p}}^{N}E_{i,f_i(k)}\sqrt{R^{\mathrm{AWG}}_{p,f_i(k)}}}_{\text{AWG intraband xtalk}},\tag{32}$$

where $R^{\mathrm{AWG}}_{p,j}$ is the optical crosstalk power ratio (smaller than 1) for AWG output port p due to the interference from AWG output port j for signals on identical wavelengths.

The demux interband crosstalk on wavelength i due to interference from wavelength m is given by $R^D_{i,m}$, the optical crosstalk power ratio (smaller than 1). At the receiver for λ_i at switch output port p, the total field is

$$\underbrace{E_{i,f_i(p)}}_{\text{signal}}+\underbrace{\sum_{\substack{m=1\\m\neq i}}^{N}E_{m,f_m(p)}\sqrt{R^D_{i,m}}}_{\text{Demux interband xtalk}}+\underbrace{\sum_{\substack{k=1\\k\neq p}}^{N}E_{i,f_i(k)}\sqrt{R^{\mathrm{AWG}}_{p,f_i(k)}}}_{\text{AWG intraband xtalk}}.\tag{33}$$

We note that the concatenation of AWG, demux, and mux narrows the effective optical filter bandwidth [30]. In practice, these devices should be designed with a flat passband to minimize the signal degradation due to intersymbol interference (ISI), although there is a trade-off between passband flatness and insertion loss. We assume in the following analysis that all the optical filters have flat passband and that the filter narrowing effects on the signals are negligible.

Which terms of Eq. (33) will be dominant at photodetection? The beating of intraband noise with itself goes as R^{AWG}, while the beating of intraband noise with the signal goes as the square root of R^{AWG}; hence we neglect the intraband noise self-beat terms. Beating of the interband noise with the signal and with the intraband noise will fall out of the electrical bandwidth of the receiver; hence these terms are neglected. Therefore the photocurrent will be proportional to

$$E^2_{i,f_i(p)}+\sum_{\substack{m=1\\m\neq i}}^{N}E^2_{m,f_m(p)}R^D_{i,m}$$
$$+2\sum_{\substack{k=1\\k\neq p}}^{N}E_{i,f_i(k)}E^*_{i,f_i(p)}\sqrt{R^{\mathrm{AWG}}_{p,f_i(k)}}.\tag{34}$$

We assume an intensity modulated/direct detection system using OOK with equiprobable data bits $b_{i,f_i(p)}\in\{0,1\}$, and uniformly distributed random phase $\theta_{i,f(p)}\in[0,2\pi)$. The optical fields at time t can be written as

$$E_{i,f(p)}(t)=b_{i,f_i(p)}E\cdot e^{j\theta_{i,f_i(p)}(t)}.\tag{35}$$

We assume the optical power of all channels is equal for a mark (E^2), and is zero for "spaces". We make the worst-case assumption of all signals having the same polarization. We assume that the complete optical spectrum falls in the receiver passband. From Eq. (34), and assuming uniform responsivity, we can express the photocurrent for λ_i at switch output port p as

$$I_{i,p}=\underbrace{b^2_{i,f_i(p)}E^2}_{\text{signal}}+\underbrace{\sum_{\substack{m=1\\m\neq i}}^{N}b^2_{m,f_m(p)}E^2R^D_{i,m}}_{\text{demux interband xtalk}}$$
$$\underbrace{+2\sum_{\substack{k=1\\k\neq p}}^{N}b_{i,f_i(p)}b_{i,f_i(k)}E^2\sqrt{R^{\mathrm{AWG}}_{p,f_i(k)}}\cos\left(\theta_{i,f_i(p)}-\theta_{i,f_i(k)}\right)}_{\text{AWG intraband xtalk}}.\tag{36}$$

We specify the noise contribution from each component separately, to facilitate analysis in the main body of this paper. We begin by averaging over the crosstalk bits. The demux interband crosstalk bits are independent; hence the mean interband crosstalk term is reduced by one half. The beating bits in the AWG intraband crosstalk are also independent; therefore the average current from intraband crosstalk is reduced by one half. Next, we average over phase. We assume the phase of each input wavelength to be uncorrelated as N^2 independent light sources are used; hence the average intraband crosstalk is reduced by another one half due to the square of the cosine term in Eq. (36). Therefore, we have the power of the photocurrent as

$$P_{i,p} = \underbrace{b^4_{i,f_i(p)}E^4}_{\text{signal}} + \underbrace{\frac{1}{2}\sum_{\substack{m=1\\m\neq i}}^{N} E^4\left(R^D_{i,m}\right)^2}_{\text{Demux interband xtalk}} + \underbrace{b^2_{i,f_i(p)}\sum_{\substack{k=1\\k\neq p}}^{N} E^4 R^{\text{AWG}}_{p,f_i(k)}}_{\text{AWG intraband xtalk beating term}} . \tag{37}$$

We preserve $b_{i,f_i(p)}$ in Eq. (37) to properly capture their noise contribution when conditioning on "marks" and "spaces."

Let the adjacent (R_{AX}) and non-adjacent (R_{NX}) crosstalk power ratios be defined as in Appendix A. Then

$$R^{\text{AWG}}_{p,j} = \begin{cases} R_{AX} & |p-j| \equiv 1 \text{ or } N-1 \ (\text{mod } N) \\ R_{NX} & \text{otherwise.} \end{cases} \tag{38}$$

A similar definition, for R_{D-AX} and R_{D-NX}, holds for the demux. For full load, $L = 1$, the AWG crosstalk noise power is [16,29]

$$\sigma^2_{\text{AWG}} = E^4 \begin{cases} 2R_{AX} + (N-3)R_{NX} & b_{i,fi(p)} = 1 \\ \sim 0 & b_{i,fi(p)} = 0, \end{cases} \tag{39}$$

and the demux crosstalk noise power is

$$\sigma^2_D = E^4\left(R^2_{D-AX} + \frac{N-3}{2}R^2_{D-NX}\right). \tag{40}$$

When $L \le 1$, for a given output port, the crosstalk seen by any given wavelength can come from any of $N-1$ input ports, each with probability L. Since we consider only the intraband crosstalk in the AWG, we can use the same analysis as in Appendix A with $\rho = L$. Thus, for a mark, we again have

$$\sigma^2_{\text{AWG}}\left(i_{AX}, i\right) = E^4\left[i_{AX}R_{AX} + \left(i - i_{AX}\right)R_{NX}\right]. \tag{41}$$

The expected value of the intraband crosstalk for a multiple wavelength per port AWG of size $N \times N$ and effective offered load $\rho = L$ is

$$\sigma^2_{\text{AWG}} = E\left[\sigma^2_{\text{AWG}}\left(i_{AX}, i\right)\right] = X(N, LN), \tag{42}$$

and for the demux

$$\begin{aligned} X_D(N,\rho) &= \mathrm{E}\sigma^2_D = E^4 \cdot \mathrm{E}\left[i_{AX}R^2_{D-AX} + \frac{i-i_X}{2}R^2_{D-NX}\right] \\ &= E^4 R^2_{D-NX}\rho(N-1) \\ &\quad + E^4\left(R^2_{D-AX} - R^2_{D-NX}\right)\sum_{i=1}^{N-1}\text{PMF}(i) \\ &\quad \times \left[P_X(1,i) + 2P_X(2,i)\right]. \end{aligned} \tag{43}$$

Appendix C: ASE Contribution From EDFAs and TWCs

The ASE power for an EDFA is [29]

$$P_{EDFA-ASE} = 2n_{sp}hv_0 \cdot (G-1)\cdot \Delta v_r, \tag{44}$$

where the factor of 2 accounts for the unpolarized nature of the ASE, n_{sp} is the spontaneous emission factor, h is Planck's constant, v_0 is the carrier frequency in Hz, i.e., 193.5 THz for wavelength around 1550 nm, G is the EDFA gain, and λv_r is the optical filter bandwidth.

For the sake of simplicity, we consider a balanced chain. We consider a typical low-noise EDFA with a noise figure $NF = 2n_{sp}(G-1)/G \approx 2n_{sp} = 4$ dB, and $G = 18$ dB. Using Eq. (44), the ASE power spectrum at each EDFA is −36 dBm/0.1 nm. This ASE sees no net amplification in the balanced chain.

The TWC ASE noise spectrum is −41 dBm/0.1 nm; however, this noise sees a net gain of 10 dB in the balanced chain (the consequence of balancing the 10 dB of conversion loss in the signal). To summarize, we use the following ASE contributions:

$$\begin{cases} ASE_{TWC} = -31 \text{ dBm/0.1 nm} \\ ASE_{EDFA} = -36 \text{ dBm/0.1 nm.} \end{cases} \tag{45}$$

We assumed an electrical filter bandwidth at 70% of the transmission rate; optical filtering is twice the bandwidth of the electrical filter. Spontaneous–spontaneous beating is neglected. For a unit responsivity photodetector, the noise due to the ASE is [29]

$$\begin{cases} \sigma^2_{\text{ASE-EDFA}} = -31 \text{ dBm} \cdot \dfrac{2\times 0.7R_S}{12.5 \text{ GHz}} \\ \sigma^2_{\text{ASE-TWC}} = -36 \text{ dBm} \cdot \dfrac{2\times 0.7R_S}{12.5 \text{ GHz}}, \end{cases} \tag{46}$$

where R_S is the transmission rate and 12.5 GHz is the equivalent frequency bandwidth for a wavelength window width of approximately 0.1 nm.

Acknowledgments

This work was supported by the Canadian Institute for Photonic Innovation (CIPI). Yousra Ben M'Sallem acknowledges the Natural Sciences and Engineering Research Council (NSERC) of Canada for a graduate scholarship. This work was presented in part at the *IEEE High Performance Switching and Routing Conference* in July 2011.

References

[1] R. W. Tkach, "Scaling optical communications for the next decade and beyond," *Bell Labs Technical J.*, vol. 14, pp. 3–9, Win 2010.

[2] L. A. Barroso and U. Hölzle, "The datacenter as a computer: An introduction to the design of warehouse-scale machines," in *Synthesis Lectures on Computer Architecture*. Morgan and Claypool Publishers, 2009, p. 108.

[3] A. F. Benner, P. K. Pepeljugoski, and R. J. Recio, "A roadmap to 100G ethernet at the Enterprise data center," *IEEE Commun. Mag.*, vol. 45, pp. 10–17, Nov. 2007.

[4] A. Leon-Garcia, K. Khavari, H. Rastegarfar, and A. Bigdeli, “Optical switching and scalability in datacenters,” presented at the 2010 IEEE Photonics Society Summer Topical Meeting, Playa del Carmen, 2010.

[5] X. Ye, Y. Yin, D. Ding, S. Johnson, V. Akella, and S. J. B. Yoo, “Assessment of optical switching in data center networks,” presented at the Optical Fiber Communication (OFC), collocated National Fiber Optic Engineers Conf., San Diego, CA, 2010.

[6] S. J. Ben Yoo, “Optical packet and burst switching technologies for the future photonic Internet,” *J. Lightwave Technol.*, vol. 24, pp. 4468–4492, Dec. 2006.

[7] O. Ishida and H. Takahashi, “Loss-imbalance equalization in arrayed-wave-guide-grating (Awg) multiplexer cascades,” *J. Lightwave Technol.*, vol. 13, pp. 1155–1163, June 1995.

[8] K. Okamoto, T. Hasegawa, O. Ishida, A. Himeno, and Y. Ohmori, “32×32 arrayed-waveguide grating multiplexer with uniform loss and cyclic frequency characteristics,” *Electron. Lett.*, vol. 33, pp. 1865–1866, Oct. 23, 1997.

[9] G. Contestabile, L. Banchi, M. Presi, and E. Ciaramella, “Investigation of transparency of FWM in SOA to advanced modulation formats involving intensity, phase, and polarization multiplexing,” *J. Lightwave Technol.*, vol. 27, pp. 4256–4261, Oct. 1, 2009.

[10] Z. H. Li, Y. Dong, J. Y. Mo, Y. X. Wang, and C. Lu, “Cascaded all-optical wavelength conversion for RZ-DPSK signal based on four-wave mixing in semiconductor optical amplifier,” *IEEE Photon. Technol. Lett.*, vol. 16, pp. 1685–1687, July 2004.

[11] M. Matsuura, N. Kishi, and T. Miki, “Ultrawideband wavelength conversion using cascaded SOA-based wavelength converters,” *J. Lightwave Technol.*, vol. 25, pp. 38–45, Jan. 2007.

[12] D. Nielsen, S. L. Chuang, N. J. Kim, D. Lee, S. H. Pyun, W. G. Jeong, C. Y. Chen, and T. S. Lay, “High-speed wavelength conversion in quantum dot and quantum well semiconductor optical amplifiers,” *Appl. Phys. Lett.*, vol. 92, May 26, 2008.

[13] C. Politi, D. Klonidis, and M. J. O’Mahony, “Waveband converters based on four-wave mixing in SOAs,” *J. Lightwave Technol.*, vol. 24, pp. 1203–1217, Mar. 2006.

[14] Q. Xu, A. Ghazisaeidi, S. Doucet, Y. Ben M’Sallem, L. A. Rusch, and S. LaRochelle, “Tunable SOA wavelength converter for optical packet switching router,” presented at the IEEE Photonics Society 23rd Ann. Meeting, Denver, CO, 2010.

[15] Q. Xu, H. Rastegarfar, Y. Ben M’Sallem, S. LaRochelle, A. Leon-Garcia, and L. A. Rusch, “OOK Q-factor degradation in scalable optical switches,” presented at the IEEE Int. Conf. on High Performance Switching and Routing, Cartagena, Spain, 2011.

[16] R. Gaudino, G. A. G. Castillo, F. Neri, and J. M. Finochietto, “Can simple optical switching fabrics scale to terabit per second switch capacities?,” *J. Opt. Commun. Netw.*, vol. 1, pp. B56–B69, Aug. 1, 2009.

[17] S. Kamei, M. Ishii, A. Kaneko, T. Shibata, and M. Itoh, “$N\times N$ cyclic-frequency router with improved performance based on arrayed-waveguide grating,” *J. Lightwave Technol.*, vol. 27, pp. 4097–4104, Sept. 15, 2009.

[18] M. C. Chia, D. K. Hunter, I. Andonovic, P. Ball, I. Wright, S. P. Ferguson, K. M. Guild, and M. J. O’Mahony, “Packet loss and delay performance of feedback and feed-forward arrayed-waveguide gratings-based optical packet switches with WDM inputs–outputs,” *J. Lightwave Technol.*, vol. 19, no. 9, pp. 1241–1254, 2001.

[19] L. Li, S. D. Scott, and J. S. Deogun, “A novel fiber delay line buffering architecture for optical packet switching,” presented at the IEEE Global Telecommunications Conf., 2003.

[20] S. Iyer, R. Zhang, and N. McKeown, “Routers with a single stage of buffering,” *ACM SIGCOMM Computer Communications Review*, vol. 32, pp. 251–264, 2002.

[21] C. Clos, “A study of non-blocking switching networks,” *Bell System Technical J.*, vol. 32, pp. 406–424, 1953.

[22] J. Cheyns, C. Develder, E. Van Breusegem, D. Colle, F. De Turck, P. Lagasse, M. Pickavet, and P. Demeester, “Clos lives on in optical packet switching,” *IEEE Commun. Mag.*, vol. 42, pp. 114–121, Feb. 2004.

[23] H. Takahashi, K. Oda, and H. Toba, “Impact of crosstalk in an arrayed-waveguide multiplexer on $N\times N$ optical interconnection,” *J. Lightwave Technol.*, vol. 14, pp. 1097–1105, June 1996.

[24] S. J. Ben Yoo, F. Xue, Y. Bansal, J. Taylor, Z. Pan, J. Cao, M. Jeon, T. Nady, G. Goncher, K. Boyer, K. Okamoto, S. Kamei, and V. Akella, “High-performance optical-label switching packet routers and smart edge routers for the next-generation internet,” *IEEE J. Sel. Areas Commun.*, vol. 21, pp. 1041–1051, Sept. 2003.

[25] M. R. Hashemi and A. LeonGarcia, “A general purpose cell sequencer/scheduler for ATM switches,” in *IEEE Infocom’97 - the Conf. on Computer Communications, Proc.*, 1997, vol. 1–3, pp. 29–37.

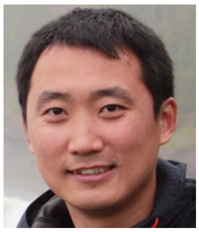
[26] Y. K. Yeo, Z. W. Xu, D. W. Wang, J. G. Liu, Y. X. Wang, and T. H. Cheng, “High-speed optical switch fabrics with large port count,” *Opt. Express*, vol. 17, pp. 10990–10997, June 22, 2009.

[27] JDSU, “100 GHz Wavelength Division Multiplexer/Demultiplexer” [Online]. Available: http://www.jdsu.com.

[28] Enablence, “100 GHz $N\times N$ Arrayed Waveguide Router” [Online]. Available: http://www.enablence.com.

[29] G. P. Agrawal, *Fiber-optic Communication Systems*. 3rd ed. Wiley-Interscience, New York, 2002.

[30] I. Roudas, N. Antoniades, T. Otani, T. E. Stern, R. E. Wagner, and D. Q. Chowdhury, “Accurate modeling of optical multiplexer/demultiplexer concatenation in transparent multiwavelength optical networks,” *J. Lightwave Technol.*, vol. 20, pp. 921–936, June 2002.



Qing Xu received his B.E. degree in information engineering from Jiaotong University (Shanghai, P.R. China) in 2002, and his diplôme d’ingénieur of telecommunications from Ecole Spéciale de Mécanique et d’Electricité (Paris, France) in 2005, respectively. He received his Ph.D. degree in electrical engineering from Ecole Nationale Supérieure des Télécommunications (Paris, France) in April 2009, where he conducted researches on experimental quantum key distribution system and optical coherent detection technologies.

He worked as postdoctoral researcher in the Centre d’Optique, Photonique et Laser in Université Laval, Québec, Canada, from 2009–2011, where he conducted research on optical packet switching technologies. He is author or co-author of over 20 international technical papers.

He was a CDMA R&D Engineer in Wide Telecom Inc. from 2002 to 2003. Since 2012 he is a researcher in Ericsson Inc. (Montreal, Canada) where he conducts research on optical interconnects technologies.

Houman Rastegar was born in Tehran, Iran, in 1983. He received B.Sc. degrees in Electrical Engineering (Communications) and Computer Engineering (Software) from Amirkabir University of Technology (Tehran Polytechnic), Tehran, Iran, in 2006 and 2008, respectively. He received his M.Sc. degree in Microwave and Optical Communications from Sharif University of Technology, Tehran, Iran, in 2008. Since September 2008 he has been a Ph.D. student in electrical engineering at the Network Architecture and Services Group of University of Toronto. His research interests include optical networks, WDM transmission technologies and systems, optical buffering, and optical switch architectures. He is a student member of IEEE Photonics Society, OSA, and SPIE.

Yousra BenM'Sallem was born in Tunis, Tunisia in 1982. She received her telecommunications engineer diploma from Sup'Com (Ecole Supérieure de Communications), Tunis, Tunisia, in 2006.

She received her M.Eng. degree in electrical engineering from Université Laval in 2008. She is currently working toward the Ph.D. degree in optical packet switched networks (OPSNs) at the Center for Optics, Photonics and Lasers, Department of Electrical and Computer Engineering, Université Laval. Ms. Ben M'Sallem is the recipient of a prestigious Alexander Graham Bell Canada Graduate Scholarship (2010) from the National Sciences and Engineering Research Council of Canada (NSERC). She is a student member of the IEEE Photonics Society and the Optical Society of America (OSA) and the current treasurer of Jeux Photoniques and Canadian Institute for Photonic Innovations for Students (CIPI-S).

Al Leon-Garcia is a professor of electrical and computer engineering at the University of Toronto. He is a Fellow of the Engineering Institute of Canada. He received the 2006 Thomas Eadie Medal from the Royal Society of Canada and the 2010 IEEE Canada A. G. L. McNaughton Gold Medal for his contributions to the area of communications. He holds a Canada Research Chair in Autonomic Service Architecture. His current research interests are focused on application-oriented networking and autonomic resource management with a focus on enabling pervasive smart infrastructure. His research team is currently developing a network and applications testbed that supports at-scale experimentation in new network protocols and distributed applications.

Sophie LaRochelle received a Bachelor's degree in engineering physics from Université Laval, Canada, in 1987; and a Ph.D. degree in optics from the University of Arizona, USA, in 1992.

From 1992 to 1996, she was a Research Scientist at the Defense Research and Development Canada - Valcartier, where she worked on electro-optical systems. She is now a professor at the Department of Electrical and Computer Engineering, Université Laval. Her current research activities are focused on active and passive components for optical communication systems including fiber Bragg gratings, optical amplifiers, Silicon photonic devices, multi-wavelength and pulsed lasers. Other research interests include packet-switched networks with photonic code processing and transmission of radio-over-fiber signals including UWB and GPS. Dr. LaRochelle is a member of OSA and of IEEE-PS.

Leslie Ann Rusch received the B.S.E.E. degree (with honors) from the California Institute of Technology, Pasadena, in 1980 and the M.A. and Ph.D. degrees in electrical engineering from Princeton University, Princeton, NJ, in 1992 and 1994, respectively. Dr. Rusch has experience in defense, industrial and academic communications research. She was a communications project engineer for the Department of Defense from 1980–1990. While on leave from Université Laval she spent two years (2001–2002) at Intel Corporation. She is currently a Professor in the Department of Electrical and Computer Engineering at Université Laval, QC, Canada, performing research on wireless and optical communications. Prof. Rusch's research interests include advanced optical networks exploiting all optical wavelength conversion, analysis of phase noise in optical systems using coherent detection, semiconductor and erbium-doped optical amplifiers and their dynamics; and in wireless communications, optimization of the optical/wireless interface in emerging cloud based computing networks, optical pulse shaping for high-bit rate ultrawide-band (UWB) systems, and implantable medical sensors with high bit rate UWB telemetry. She is currently an associate editor for the *OSA/IEEE Journal of Optical Communications and Networking*.